
GRIN (GRADIENT-INDEX) OPTICS

Numerical Integrators and Lenses

SIMULATIONS

SEDSCelestia

Geometric and gradient-index optics in two dimensions.

Integrator analysis, conservation laws, lens design,
and optimization.

Core Member Induction

Due: EOD, 18th July 2026

Prerequisites: Basic Python, introductory optics and a curious mind.

This assignment is split and arranged in tasks of increasing difficulty. You are required to complete all the tasks before the deadline and submit the Google Form with your GitHub repo included in it. In case of inability to complete, you may still submit as genuine effort and thinking might be considered.

INTRODUCTION.

Gradient Index (GRIN) Optics is a fascinating field that explores the behavior of light in media where the refractive index varies spatially. Unlike traditional optics, where light travels in straight lines through homogeneous materials, GRIN optics allows for the manipulation of light paths by engineering the refractive index distribution. This leads to a wide range of applications, from compact lens design to advanced imaging systems.

In this assignment, we will delve into the principles of GRIN optics, analyze the behavior of light, methods to simulate the light path accurately using various numerical integrators, and design a few GRIN lenses, resulting in familiarizing you with the simulation of particle trajectories under the interesting laws of optics. The tasks are designed to progressively build your understanding of integrating the particle's trajectories, analyzing the integrators, and then applying that understanding to unique path design problems by manipulating the refractive index distribution.

GRIN Optics Basics.

The rays in GRIN media are governed by these two differential equations:

$$\frac{dy}{ds} = \frac{p}{n(y)} \quad (1)$$

$$\frac{dp}{ds} = \nabla n(y) \quad (2)$$

Where s is the path length along the ray, y is the position of the ray, p is the momentum of the ray, and $n(y)$ is the refractive index at position y . The first equation describes how the position changes with respect to the path length, while the second equation describes how the momentum changes due to the gradient of the refractive index.

I. LINEAR GRIN MEDIA AND INTEGRATOR ANALYSIS.

You have to build a simulation of ray trajectories in a linear GRIN medium, where the refractive index is given by the equation:

$$n(y) = n_0 + \alpha y$$

The setup:

There'll be a state variable that holds the position and momentum of the particle.

$$\mathbf{S} = \begin{bmatrix} x \\ y \\ p_x \\ p_y \end{bmatrix}$$

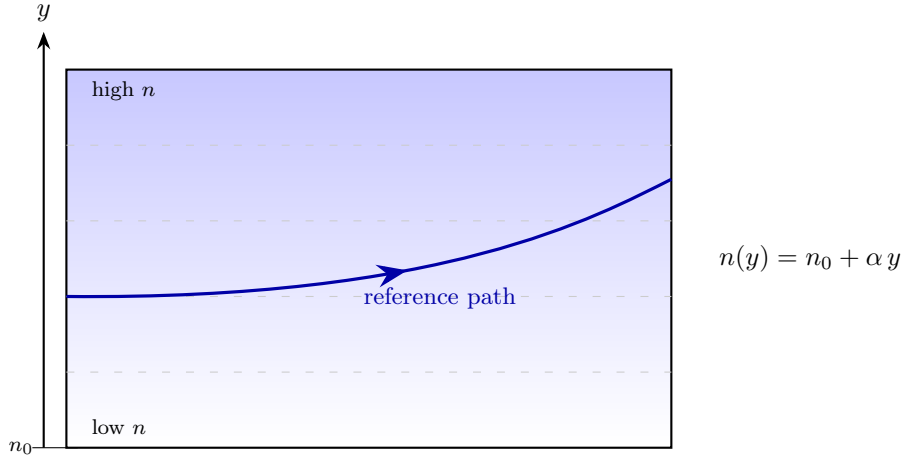


Figure 1: You are expected to have similar setup for the simulation of the particle. the refractive index is dependent on the y-coordinate. Increasing the optical density of the medium as we go up. The entire simulation box is filled with the medium. The expected path is a parabola which steeps more aggressively as α increases.

Part A. Simulating the Trajectory.

Simulate the trajectory of the particle using the following numerical integrators:

Euler Method. The simplest numerical method for solving ordinary differential equations (ODEs). It approximates the solution by taking small steps along the direction of the derivative. However, it can be inaccurate and unstable for stiff equations or long time intervals.

$$\mathbf{S}_{n+1} = \mathbf{S}_n + f(\mathbf{S}_n)\Delta s$$

Runge-Kutta 4th Order (RK4). A more accurate method that considers intermediate slopes to achieve better accuracy. It calculates four estimates of the slope at different points within the interval and combines them to produce a weighted average slope.

$$\begin{aligned}
k_1 &= f(\mathbf{S}_n) \\
k_2 &= f\left(\mathbf{S}_n + \frac{\Delta s}{2} k_1\right) \\
k_3 &= f\left(\mathbf{S}_n + \frac{\Delta s}{2} k_2\right) \\
k_4 &= f(\mathbf{S}_n + \Delta s k_3) \\
\mathbf{S}_{n+1} &= \mathbf{S}_n + \frac{\Delta s}{6} (k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}$$

Open Exploration. You are required to find and implement one integrator of your choice. Higher order Runge-Kutta methods, Verlet integration, Leapfrog method, or any other integrator that you find interesting and suitable for this problem.

Part B. Analyzing Numerical Integrators.

For this part, you are required to analyze the accuracy and stability of the integrators (euler, RK4, open-exploration one) you implemented. You have to test it against the analytical solution of the ray trajectory. The true trajectory (analytical solution) for a particle with initial y-coordinate y_0 and angle from the horizontal θ_0 is given by the equation:

$$y_{\text{true}}(x) = \frac{(n(y_0) \cos \theta_0) \cdot \cosh\left(\frac{\alpha x}{p} + \operatorname{arctanh}(\sin \theta_0)\right) - n_0}{\alpha}$$

For a horizontal starting angle ($\theta_0 = 0$), it simplifies to:

$$y_{\text{true}}(x) = \frac{p \cosh\left(\frac{\alpha x}{p}\right) - n_0}{\alpha}$$

But for small values of α , the trajectory can be approximated as a parabola.

For analyzing the integrators, the accuracy can be determined by lower values of RMSE.

$$\epsilon_{\text{traj}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{\text{num}}(x_i) - y_{\text{true}}(x_i))^2}$$

You'll be provided a utility python script to calculate the RMSE and Compute so you don't have to bother with the unnecessary stuffs. You should submit a scatter plot between Compute and Accuracy in your submission repository and conclude which integrator you think is better suited for this simulation and provide reasoning for the same.

The parameter for compute will be the number of derivative evaluations. For example, RK-4 has 6 times more derivative evaluations than Euler's method.

II. MAXWELL'S FISH-EYE LENS.

The Fish-Eye Lens is used to create a perfect image of a point source at the opposite side of the lens. In simple terms, light rays emanating from a point source will be focused to a single point on the opposite side of the lens, creating a perfect image. The refractive index distribution for Maxwell's Fish-Eye Lens is given by the equation:

$$n(r) = \frac{n_0}{1 + \left(\frac{r}{a}\right)^2}$$

Where $r = \sqrt{x^2 + y^2}$

$$\nabla n(r) = -\frac{2n_0}{a^2 \left(1 + \left(\frac{r}{a}\right)^2\right)^2} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

The scenario:

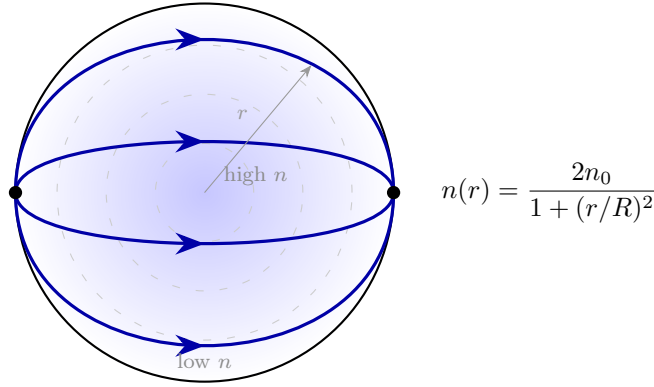


Figure 2: The light rays emerging in all directions inside the lens converge at the same point.

Simulate the ray trajectories in Maxwell's Fish-Eye Lens and verify that all rays emanating from a point source at one focal point converge to the opposite focal point. You can use any of the integrators you implemented in the previous section for this simulation.

Bouguer's Invariant. (Optical Angular Momentum). Bouguer's Invariant is a principle in optics for spherically symmetric media. It states that the product of the radial distance, the refractive index, and the sine of the angle between the ray direction and the radial vector is constant along a ray path. Basically, it is a direct statement of the conservation of *optical* angular momentum. Mathematically,

$$\mathcal{L} = xp_y - yp_x = \text{constant}$$

Calculate this value $\mathcal{L}$ at every step and make sure it remains constant or *almost* constant.

III. LUNEBURG LENS.

The Luneburg Lens is a spherically symmetric gradient-index lens that focuses light from any direction onto a point on the opposite side of the lens. The refractive index distribution for a Luneburg Lens with radius a is given by the equation:

$$n(r) = \sqrt{2 - \left(\frac{r}{a}\right)^2}$$

Where $r = \sqrt{x^2 + y^2}$

$$\nabla n(r) = \frac{-r}{a^2 \cdot n(r)}$$

The scenario:

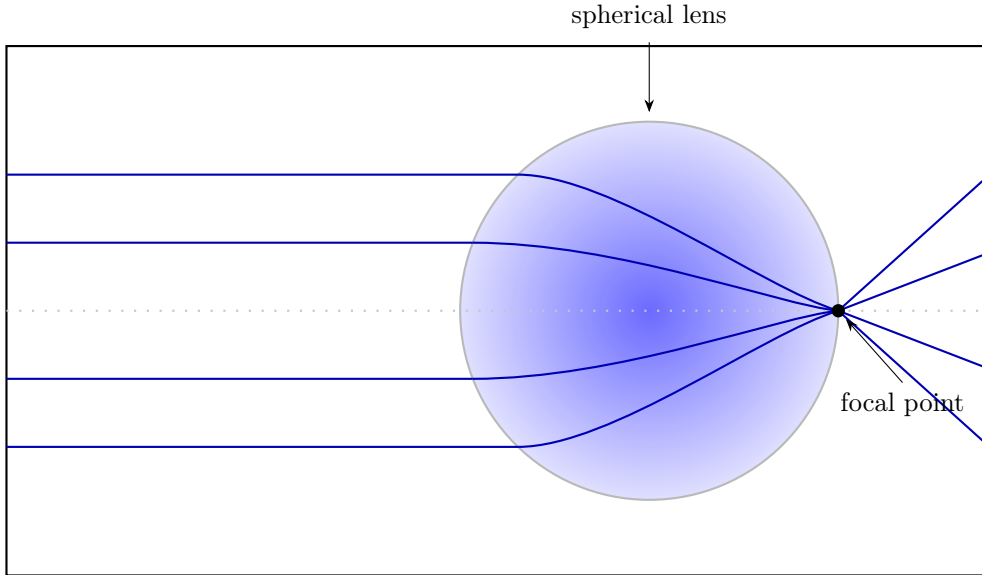


Figure 3: Outside the lens, the refractive index is uniform. Inside the lens, the refractive index is given by the above equation. (The bending of light is innacurately exaggerated in the figure for visual clarity.)

Simulate the ray trajectories in a Luneburg Lens and verify that all rays emanating from a point source at one focal point converge to the opposite focal point. You can use any of the integrators you implemented in the previous section for this simulation.

Verifying the Focusing Property. To numerically quantify the focusing property of the lens, you can calculate the variance in the final positions of the rays on the opposite side of the lens. You can define a horizontal line along the center of the lens and calculate the variance of the y-coordinates of the rays when they intersect this line. A smaller variance indicates better focusing.

Resources.

- [Gradient-Index Optics - Visual Introduction](#)
- [Gradient-Index Optics - Wikipedia](#)